

Example Sheet: Mode shapes

Exercise 1.

A 2-DOF vibration system is described by the matrix equations of motion

$$M \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} + K \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

where the mass matrix M and stiffness matrix K are given by

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ kg}, \quad K = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \text{ Nm}^{-1}.$$

Verify that $\omega_1 = \sqrt{2} \text{ rad/s}$ and $\omega_2 = 2 \text{ rad/s}$ are natural frequencies of the system, and determine and sketch the associated mode shapes. Comment on the number of nodes in each mode.

Solution 1.

For the 2 DOF system given in the question, the matrix equation of motion is of the form

$$M \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} + K \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

Using the given mass and stiffness matrices M and K , the natural frequencies are the square roots of the eigenvalues of the matrix $A = M^{-1}K$. Since $M = I$ we have

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

We verify if the given frequencies are natural frequencies by trying to determine the mode shapes, i.e. the eigenvectors of A . Noting that $\lambda_1 = \omega_1^2 = 2$ and $\lambda_2 = \omega_2^2 = 4$, the eigenvector equations are

$$\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

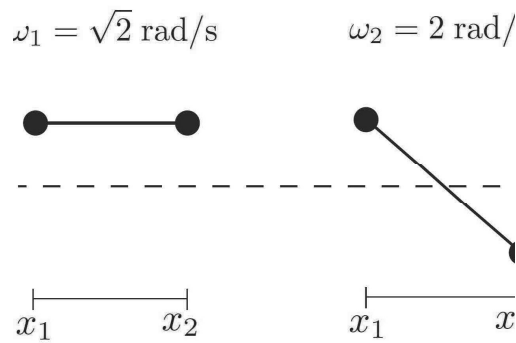
and

$$\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 4 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

yielding normalised eigenvectors of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, respectively.

You should check consistency of the two equations generated for each mode to verify that ω_1^2 and ω_2^2 are indeed eigenvalues.

Sketches of the two modes similar to those below show that ω_1 (fundamental) mode has no nodes, while ω_2 mode has a single node in between x_1 and x_2 .

**Exercise 2.**

A drivetrain is effectively modelled as a free-free system with the following equation of motion:

$$\begin{pmatrix} I_1 & 0 & 0 \\ 0 & 2I_2 & 0 \\ 0 & 0 & I_1 \end{pmatrix} \begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{\theta}_3 \end{pmatrix} + \begin{pmatrix} k_1 & -k_1 & 0 \\ -k_1 & 2k_1 & -k_1 \\ 0 & -k_1 & k_1 \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (1)$$

- The drive-train has a natural mode with angular frequency $\omega = 0$. Describe the motion of the inertias associated with this mode. *Note: This does not require solving the equations of motion.*
- It has been reported that the system modeled by equation (1) can freely vibrate with a frequency of $\omega_1 = \sqrt{\frac{k_1}{I_1}}$. Demonstrate whether the model is consistent with this report and give the relative motions of the equivalent inertias at this frequency.
- Do you expect the system to have further modes of free vibration? If yes, qualitatively sketch the mode shape(s). How would the frequency of the mode(s) compare qualitatively to ω_1 ?

Solution 2.

- Note: One can easily verify that the equation given in the question describes the equivalent system for the values given.* The mode with angular frequency $\omega = 0$ corresponds to the free-wheel mode; i.e. all equivalent inertias rotate with the same angular velocity. This is a fundamental property of all free-free systems.
- Assuming modes of free vibration of the form $\theta_i = A_i \sin \omega_n t, i = 1, 2, 3$ one can substitute into the three model equations for $\omega_1 = \sqrt{\frac{k_1}{I_1}}$:

$$\begin{aligned} -\omega_1^2 I_1 A_1 + k_1 A_1 - k_1 A_2 &= 0 & \Rightarrow k_1 A_2 &= 0 & A_2 &= 0 \\ -2\omega_1^2 I_2 A_2 - k_1 A_1 + 2k_1 A_2 - k_1 A_3 &= 0 & \Rightarrow -k_1 A_1 - k_1 A_3 &= 0 & A_1 &= -A_3 \\ -\omega_1^2 I_1 A_3 - k_1 A_2 + k_1 A_3 &= 0 & \Rightarrow k_1 A_2 &= 0 \end{aligned}$$

So ω_1 is a solution, where I_1 and I_3 move equally in opposite directions, while I_2 is at rest.

- iii) As the system has three degrees of freedom we expect three modes of free vibration. So far, we found two with frequencies $0, \sqrt{\frac{k_1}{I_1}}$ with zero and one nodes, so we are missing one more mode.

This mode must have two nodes, so one between each pair of inertias: The first and the last move in the same direction, the middle one in the opposite direction.

More nodes mean higher frequency, thus $\omega_2 > \omega_1$.

Exercise 3.

A light shaft carries three inertias. In consistent units, the equations of motion for the system are

$$\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{\theta}_3 \end{pmatrix} + \begin{pmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (2)$$

From these, the two natural frequencies of torsional oscillations are $\omega_n^2 = 3 \pm \sqrt{3}$.

Find the mode shapes associated with these natural frequencies, normalising to give unit amplitude for θ_1 . Comment briefly on the number of modes that this system can support.

Solution 3.

Simplest to note that eigenvectors give the mode shapes and eigenvalues are ω_n^2 . The set up standard for of the eigenvector definition, setting A_1 to 1 as required.

$$\begin{pmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ A_2 \\ A_3 \end{pmatrix} = (3 \pm \sqrt{3}) \begin{pmatrix} 1 \\ A_2 \\ A_3 \end{pmatrix}.$$

Now A_2 can be read from the top row, and then A_3 found easily from the bottom row.

Mode 1: $\left(1, \frac{\sqrt{3}-1}{2}, -\frac{\sqrt{3}-1}{2(2-\sqrt{3})}\right)$. Mode 2: $\left(1, -\frac{\sqrt{3}+1}{2}, \frac{\sqrt{3}+1}{2(2+\sqrt{3})}\right)$.

Three inertias in a freely rotating system (free-free) give only two vibrations. The third mode is the rigid body rotation equivalent to $\omega_n = 0$.